

1. Differentiate between image classification, object detection, and image segmentation.

- **Image Classification:** Assigns one or more class labels to the entire image. It does not identify the exact location of objects.
- **Object Detection:** Identifies objects and determines their locations using **bounding boxes** and class labels.
- **Image Segmentation:** Classifies individual pixels of an image, providing a detailed outline or mask of each object.

Example: For an image containing a car and a person, classification may say "car and person"; detection gives bounding boxes around them; segmentation identifies the exact pixels belonging to the car and person.

2. Explain the working principle of the YOLO (You Only Look Once) object detection algorithm.

YOLO treats object detection as a single regression problem. The input image is processed by a neural network in **one forward pass**.

The basic process is:

1. The image is divided into regions/cells.
2. The network predicts bounding boxes for possible objects.
3. It predicts the **class probabilities** and **confidence scores** for these boxes.
4. Low-confidence detections are removed.
5. **Non-Maximum Suppression (NMS)** is generally used to eliminate duplicate overlapping boxes.
6. The remaining boxes and labels represent the detected objects.

Because YOLO performs detection in a single pass, it provides very fast inference and is suitable for real-time applications.

3. Compare YOLO, SSD, and Faster R-CNN based on speed, accuracy, and practical applications.

Model	Speed	Accuracy	Practical Applications
YOLO	Very fast	High	Real-time detection, surveillance, autonomous systems
SSD	Fast	High	Mobile and real-time applications
Faster R-CNN	Slower	Generally very high	Applications where detection accuracy is more important than speed

Summary: YOLO is commonly preferred for real-time applications, SSD provides a good balance between speed and accuracy, while Faster R-CNN is useful when high detection accuracy is a priority.

4. What is a bounding box? Why is it important in object detection?

A **bounding box** is a rectangular box drawn around a detected object. It is usually represented using coordinates such as the top-left and bottom-right points or center coordinates with width and height.

It is important because it:

- Shows the **location** of an object.
- Helps distinguish multiple objects in the same image.
- Provides the spatial information required for object detection.
- Allows the detected object to be displayed along with its class label and confidence score.

5. Explain the significance of confidence score and Intersection over Union (IoU).

Confidence Score:

It indicates how confident the detection model is that a particular bounding box contains an object of the predicted class. A higher confidence score generally indicates a more reliable prediction.

Intersection over Union (IoU):

IoU measures the overlap between the **predicted bounding box** and the **ground-truth bounding box**.

$$IoU = \frac{\text{Area of Intersection}}{\text{Area of Union}}$$

An IoU closer to **1** means the predicted box overlaps the ground-truth box more closely.

Both measures are important for evaluating and filtering object detection results.

6. What are the advantages of using pre-trained deep learning models over traditional image processing techniques?

Advantages include:

1. **Better detection performance** for complex objects and scenes.
2. Ability to learn important features automatically from large datasets.
3. Reduced need for manually designed features.
4. Faster development because the model has already been trained.
5. Ability to recognize multiple object classes.
6. Better adaptability to different image conditions.
7. Support for modern applications such as autonomous driving and surveillance.

The experiment specifically uses pre-trained models such as **YOLOv8, SSD, and Faster R-CNN**.

7. Why are datasets such as MS COCO and Pascal VOC widely used for object detection research?

MS COCO and Pascal VOC are widely used because they provide **standardized image datasets with annotated objects**. They allow researchers to:

- Train and test object detection models.
- Compare different detection algorithms.
- Evaluate detection accuracy using common metrics.
- Study multiple object classes and different real-world scenes.
- Reproduce and compare experimental results.

The experiment identifies **MS COCO, Pascal VOC, and Open Images Dataset** as standard datasets for object detection.

8. Mention five real-world applications of object detection in computer vision.

Five applications are:

1. **Autonomous vehicles** – detecting cars, pedestrians, traffic signs, etc.
2. **Surveillance systems** – identifying and tracking objects or people.
3. **Healthcare** – detecting relevant structures or abnormalities in medical images.

4. **Retail analytics** – monitoring products, customers, and store activity.
5. **Industrial inspection** – detecting defects or objects during manufacturing.

These applications are also consistent with the experiment's stated use cases.

9. What challenges are commonly encountered while detecting objects in complex real-world environments?

Common challenges include:

- **Occlusion**, where one object partially hides another.
- Different object **sizes and scales**.
- Poor lighting or changes in illumination.
- Complex or cluttered backgrounds.
- Objects appearing at different orientations.
- Low-resolution or blurry images.
- Similar-looking objects.
- Multiple objects appearing close together.
- False detections and missed objects.

The experiment specifically recommends analyzing **correct detections, false detections, and missed objects** when evaluating results.

10. How can object detection systems be further improved using recent advancements in deep learning and computer vision?

Object detection can be improved by:

1. Using newer and more efficient detection architectures.
2. Training on larger and more diverse datasets.
3. Applying **data augmentation** to improve robustness.
4. Improving feature extraction using advanced neural network architectures.
5. Using better loss functions and optimization techniques.
6. Improving detection of small and overlapping objects.
7. Using model compression and optimization for faster inference.
8. Combining object detection with tracking and other computer vision techniques.
9. Using improved hardware such as GPUs for faster processing.
10. Continuously evaluating models for both **accuracy and inference speed**.